

Unit 6: Limits Involving Infinity; Asymptotes — Full Solutions

Warm-up

1. $\lim_{x \rightarrow \infty} \frac{3x+5}{2x-1}$

Top degree = 1, bottom degree = 1 — same degree. Divide every term by x (the highest power in the denominator):

$$\frac{3 + \frac{5}{x}}{2 - \frac{1}{x}}$$

As $x \rightarrow \infty$, $\frac{5}{x} \rightarrow 0$ and $\frac{1}{x} \rightarrow 0$:

$$\frac{3+0}{2-0} = \frac{3}{2}$$

Answer: $\frac{3}{2}$.

2. $\lim_{x \rightarrow \infty} \frac{2x+1}{x^2+3}$

Top degree = 1, bottom degree = 2 — top is smaller. By the rule, the limit is 0.

(To see it: divide by x^2 : $\frac{\frac{2}{x} + \frac{1}{x^2}}{1 + \frac{3}{x^2}} \rightarrow \frac{0+0}{1+0} = 0$.)

Answer: 0.

3. $\lim_{x \rightarrow \infty} \frac{x^3+2}{5x^2-1}$

Top degree = 3, bottom degree = 2 — top is bigger. By the rule, this limit is $+\infty$ (grows without bound).

(To see it: divide by x^2 : $\frac{x + \frac{2}{x^2}}{5 - \frac{1}{x^2}} \rightarrow \frac{x+0}{5-0}$, and as $x \rightarrow \infty$ this keeps growing forever.)

Answer: $+\infty$ (no finite limit — the fraction grows without bound).

4. $\lim_{x \rightarrow -\infty} \frac{4x^2-3}{2x^2+7}$

Same degree top and bottom (2 and 2). Divide by x^2 :

$$\frac{4 - \frac{3}{x^2}}{2 + \frac{7}{x^2}} \rightarrow \frac{4-0}{2+0} = \frac{4}{2} = 2$$

(Since x^2 is positive whether $x \rightarrow \infty$ or $x \rightarrow -\infty$, the answer is the same either direction.)

Answer: 2.

5. $\lim_{x \rightarrow \infty} \frac{5}{x^3}$

This is exactly the basic pattern $\frac{c}{x^n} \rightarrow 0$.

Answer: 0.

6. $\lim_{x \rightarrow \infty} \frac{7x^2 - 2x + 1}{3x^2 + 4x - 5}$

Same degree top and bottom (2 and 2). Divide by x^2 :

$$\frac{7 - \frac{2}{x} + \frac{1}{x^2}}{3 + \frac{4}{x} - \frac{5}{x^2}} \rightarrow \frac{7 - 0 + 0}{3 + 0 - 0} = \frac{7}{3}$$

Answer: $\frac{7}{3}$.

Standard

7. $\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2}$

Same degree top and bottom (2 and 2). Divide by x^2 :

$$\frac{5 + \frac{8}{x} - \frac{3}{x^2}}{3 + \frac{2}{x^2}} \rightarrow \frac{5 + 0 - 0}{3 + 0} = \frac{5}{3}$$

Answer: $\frac{5}{3}$.

8. $y = \frac{2x - 7}{x + 4}$

Top degree = 1, bottom degree = 1 — same degree, so the horizontal asymptote is the ratio of leading coefficients:

$$y = \frac{2}{1} = 2$$

Answer: horizontal asymptote $y = 2$.

9. $y = \frac{3x^2 + 1}{x^2 - 5x + 6}$

Same degree (2 and 2), so the horizontal asymptote is the ratio of leading coefficients:

$$y = \frac{3}{1} = 3$$

Answer: horizontal asymptote $y = 3$.

10. $y = \frac{x + 2}{x^2 - 9}$

Top degree = 1, bottom degree = 2 — top is smaller, so the horizontal asymptote is $y = 0$.

Answer: horizontal asymptote $y = 0$.

11. $y = \frac{x+1}{x^2-4}$

Factor the denominator: $x^2 - 4 = (x-2)(x+2)$, which is zero at $x = 2$ and $x = -2$.

Check the numerator at these values: at $x = 2$, $2+1 = 3 \neq 0$; at $x = -2$, $-2+1 = -1 \neq 0$. Neither factor cancels.

Answer: vertical asymptotes at $x = 2$ and $x = -2$.

12. $y = \frac{2x}{x^2+x-6}$

Factor the denominator: we need two numbers multiplying to -6 and adding to 1 : those are 3 and -2 .

$$x^2 + x - 6 = (x+3)(x-2)$$

Zero at $x = -3$ and $x = 2$.

Check the numerator ($2x$) at these values: at $x = -3$, $2(-3) = -6 \neq 0$; at $x = 2$, $2(2) = 4 \neq 0$. Neither cancels.

Answer: vertical asymptotes at $x = -3$ and $x = 2$.

Challenge

13. $y = \frac{(x+3)(x+1)}{(x+2)(x-1)}$

Horizontal asymptote: if you multiplied everything out, the top would be $x^2 + \dots$ and the bottom would be $x^2 + \dots$ — same degree (2 and 2), both with leading coefficient 1. So the horizontal asymptote is:

$$y = \frac{1}{1} = 1$$

Vertical asymptotes: the denominator is zero at $x = -2$ and $x = 1$. Check whether either factor also makes the numerator zero:

At $x = -2$: numerator $= (-2+3)(-2+1) = (1)(-1) = -1 \neq 0$.

At $x = 1$: numerator $= (1+3)(1+1) = (4)(2) = 8 \neq 0$.

Neither cancels.

Answer: horizontal asymptote $y = 1$; vertical asymptotes at $x = -2$ and $x = 1$.

14. $y = \frac{(x-2)(x+5)}{(x-2)(x+3)}$

Notice the factor $(x-2)$ appears in both the top and bottom — it cancels!

$$y = \frac{(x-2)(x+5)}{(x-2)(x+3)} = \frac{x+5}{x+3} \quad (\text{for } x \neq 2)$$

Vertical asymptote: using the simplified form, the denominator is zero at $x = -3$. Check the (simplified) numerator there: $-3 + 5 = 2 \neq 0$, so this is a genuine vertical asymptote.

At $x = 2$: since the $(x-2)$ factor canceled, this is not a vertical asymptote — it's a removable hole in the graph instead.

Horizontal asymptote: using the simplified form $\frac{x+5}{x+3}$, same degree top and bottom, leading coefficients both 1:

$$y = 1$$

Answer: horizontal asymptote $y = 1$; vertical asymptote at $x = -3$; removable hole (not an asymptote) at $x = 2$.

15. $y = \frac{x^2 - 1}{x^2 - 4x + 3}$

Factor the top (difference of squares): $x^2 - 1 = (x-1)(x+1)$.

Factor the bottom: numbers multiplying to 3, adding to -4 : -1 and -3 .

$$x^2 - 4x + 3 = (x-1)(x-3)$$

The factor $(x-1)$ cancels:

$$y = \frac{(x-1)(x+1)}{(x-1)(x-3)} = \frac{x+1}{x-3} \quad (\text{for } x \neq 1)$$

Vertical asymptote: using the simplified form, the denominator is zero at $x = 3$. Check the numerator there: $3 + 1 = 4 \neq 0$, so this is genuine.

At $x = 1$: since $(x-1)$ canceled, this is a removable hole, not a vertical asymptote.

Horizontal asymptote: simplified form $\frac{x+1}{x-3}$, same degree, leading coefficients both 1:

$$y = 1$$

Answer: horizontal asymptote $y = 1$; vertical asymptote at $x = 3$; removable hole (not an asymptote) at $x = 1$.

16. $y = \frac{3x^2 - 12}{x^2 - x - 2}$

Factor the top: $3x^2 - 12 = 3(x^2 - 4) = 3(x-2)(x+2)$.

Factor the bottom: numbers multiplying to -2 , adding to -1 : -2 and 1 .

$$x^2 - x - 2 = (x - 2)(x + 1)$$

The factor $(x - 2)$ cancels:

$$y = \frac{3(x - 2)(x + 2)}{(x - 2)(x + 1)} = \frac{3(x + 2)}{x + 1} \quad (\text{for } x \neq 2)$$

Vertical asymptote: using the simplified form, the denominator is zero at $x = -1$. Check the numerator there: $3(-1 + 2) = 3(1) = 3 \neq 0$, so this is genuine.

At $x = 2$: since $(x - 2)$ canceled, this is a removable hole, not a vertical asymptote.

Horizontal asymptote: from the original (or simplified) form, comparing leading coefficients of the top (3) and bottom (1) of the same degree:

$$y = 3$$

Answer: horizontal asymptote $y = 3$; vertical asymptote at $x = -1$; removable hole (not an asymptote) at $x = 2$.

17. $y = \frac{1}{x - 2}$

Horizontal asymptote: top degree 0 (a constant), bottom degree 1 — top is smaller, so:

$$y = 0$$

As $x \rightarrow 2^+$ (values just a bit bigger than 2, like 2.001): $x - 2$ is a very small positive number, so $\frac{1}{x - 2}$ is a very large positive number.

$$\lim_{x \rightarrow 2^+} \frac{1}{x - 2} = +\infty$$

As $x \rightarrow 2^-$ (values just a bit smaller than 2, like 1.999): $x - 2$ is a very small negative number, so $\frac{1}{x - 2}$ is a very large negative number.

$$\lim_{x \rightarrow 2^-} \frac{1}{x - 2} = -\infty$$

Answer: horizontal asymptote $y = 0$. As $x \rightarrow 2^+$, $y \rightarrow +\infty$; as $x \rightarrow 2^-$, $y \rightarrow -\infty$. (There's a vertical asymptote at $x = 2$, and the graph shoots off in opposite directions on each side of it.)

Applied

18. $C(x) = \frac{500+3x}{x}$, find the horizontal asymptote as $x \rightarrow \infty$.

Rewrite it as two separate fractions: $C(x) = \frac{500}{x} + 3$.

As $x \rightarrow \infty$, $\frac{500}{x} \rightarrow 0$, so:

$$C(x) \rightarrow 0 + 3 = 3$$

Answer: horizontal asymptote $y = 3$. This means that as the company produces more and more units, the average cost per unit gets closer and closer to \$3 — the fixed cost of \$500 gets “spread out” over so many units that it barely affects the per-unit average anymore, leaving just the \$3 variable cost per unit.

19. $C(t) = \frac{50t}{t^2+25}$, find the horizontal asymptote as $t \rightarrow \infty$.

Top degree = 1, bottom degree = 2 — top is smaller, so the limit is 0.

Answer: horizontal asymptote $y = 0$. This means that over a very long time, the drug concentration in the bloodstream approaches 0 — the body eventually clears the drug out almost entirely.

20. $P(t) = \frac{200t+500}{t+2}$, find the horizontal asymptote as $t \rightarrow \infty$.

Same degree top and bottom (1 and 1), so the horizontal asymptote is the ratio of leading coefficients:

$$y = \frac{200}{1} = 200$$

Answer: horizontal asymptote $y = 200$. This suggests that in the very long run, the city’s population levels off and approaches about 200 thousand people, rather than growing forever.

21. $R(x) = \frac{10x}{x+5}$, find the horizontal asymptote as $x \rightarrow \infty$.

Same degree top and bottom (1 and 1):

$$y = \frac{10}{1} = 10$$

Answer: horizontal asymptote $y = 10$. This means that no matter how many resources the factory has available, the production rate can never exceed (and gets closer and closer to) 10 units per hour — there’s a built-in ceiling on how fast the factory can produce, even with unlimited resources.